Cybersecurity Threat Intelligence Dashboard

ADSF - Data Visualization Project Report

Interactive dashboard | 80,200 security incidents | 2020 - Sep 2026

Tools: Python, SQL, Excel, HTML/SVG dashboard

1. Executive summary

Security teams receive a constant stream of alerts, but leadership needs a single view of **what is attacking the organisation, where it comes from, how fast it is handled and what is still open**. This project builds an interactive Cybersecurity Threat Intelligence Dashboard over 80,200 detected security incidents recorded between January 2020 and September 2026.

Incidents grew from 7,200 in 2020 to 16,000 in 2025 (+122%). 2026 covers January to September only (year to date): its 12,600 incidents are about 9% above the same nine months of 2025. Ransomware's share of all incidents rose from 12% to 27%, and in 2026 year to date it overtook phishing (26%) as the largest single category. Operational performance improved: the mitigation rate rose from 80.0% to 90.2% and the mean time to respond fell from 3.21 to 1.74 hours. The 90% mitigation target was missed in every full year (2020 to 2025) and was reached for the first time in 2026 year to date, by a margin of only 0.2 points.

Objectives

- Show total incident volume and trend by year and month, with a year selector.
- Break incidents down by threat category and sub-type, with mitigation rate per category.
- Identify attack origin countries and the industries most targeted.
- Track the response pipeline (detected, triaged, contained, mitigated) and detection-tool performance.
- Show the status of every incident by month and severity so open risk is visible.

2. Domain and business problem

Domain: Cybersecurity / IT Security Operations (applicable to Healthcare, Finance, Education, Government, E-commerce, Energy, Manufacturing and IT sectors, which are the industries tracked in the data).

Users: SOC analysts, security managers and the CISO. A manager should be able to answer in seconds: Are attacks increasing? Which attack types matter most? Are we meeting our response targets? Which incidents are still open?

3. Dataset

Source: synthetic (self-generated). No public dataset contains a clean, multi-year, incident-level record with response times and statuses, so the data was generated with seeded Python scripts (random seeds fixed for reproducibility: 2026 for 2020 to 2024, and 2027 for the 2025 to 2026 extension). Category mix, seasonality, severity distribution, response-time improvement and attack-origin countries were chosen to resemble patterns described in public threat reports. It contains no real organisations or individuals. Conclusions therefore illustrate the dashboard method and should not be treated as real-world statistics.

Size: 81,403 raw rows (CSV) reduced to 80,200 cleaned rows; 16 original columns plus 6 derived columns. **Coverage:** 2020 to 2025 are full years; 2026 is a year-to-date extract (incidents detected 1 January to 30 September 2026) and is marked with an asterisk in the dashboard.

Data dictionary

Column	Description
Incident_ID	Unique incident identifier (INC-00001 ...)
Detected_Date	Date the incident was detected
Threat_Category / Sub_Type	Six categories (Phishing, Malware, Ransomware, DDoS, Insider Threat, Web App Attack) and 3-4 sub-types each
Severity	Critical, High, Medium, Low
Source_Country / Target_Industry	Attack origin country (12 countries) and the targeted industry (8 industries)
Attack_Origin	External or Internal
Detection_Source	Tool/source that detected it: SIEM, EDR, IDS/IPS, Threat Intel Feed, User Report, Firewall
Status	Mitigated, Contained, Under Investigation, Open
Detection_Time_Min / Response_Time_Min	Minutes from occurrence to detection / from detection to response
Affected_Systems / Data_Compromised_MB	Systems impacted and data exposed

Column	Description
Financial_Impact_USD	Estimated financial impact per incident
Analyst_Team	SOC team that handled the incident
Year, Month, Month_Name (derived)	Date parts used by the dashboard slicers
SLA_Target_Min / SLA_Met (derived)	Response target by severity (Critical 60, High 120, Medium 240, Low 480 min) and whether it was met
Is_Mitigated (derived)	1 when Status = Mitigated

4. Data preparation

The raw files were deliberately messy, as real exports are. All cleaning was done in Python (pandas) and the same rules are documented here. The raw file is kept untouched in Raw_Dataset. The 2025 to 2026 records arrived as a second raw batch and went through exactly the same rules; imputation medians are taken within each batch, so the figures already published for 2020 to 2024 did not change.

Issue found in raw data	Rows affected	Treatment
Exact duplicate records	1,203	Removed
Inconsistent category labels (e.g. 'phishing ', 'PHISING', 'DDOS', 'Ransom-ware')	5,588	Trimmed, lower-cased, mapped to the six standard names
Mixed date formats (YYYY-MM-DD and DD/MM/YYYY)	about 20%	Parsed both formats; zero unparseable dates
Financial impact stored as text with '$' and commas	about 30%	Stripped symbols, converted to numeric
Impossible response times (negative or 99,999 min)	662	Set to blank, then imputed with the median for the same category and severity
Missing severity	1,184	Filled with the most common severity of the same category
Missing source country	1,577	Labelled 'Unknown' and excluded from country charts
Missing industry	824	Labelled 'Unknown' and excluded from industry charts
Detection source typed in lower case	about 1%	Standardised to official tool names

5. KPIs and definitions

KPI	Definition
Threat Statistics (big number)	Count of incidents detected in the selected year
Mitigation rate (centre ring)	Incidents with Status = Mitigated divided by incidents detected
Mitigation target	90% (assumption set in the workbook, editable)
Mean time to respond	Average Response_Time_Min, shown in hours and by month
SLA met %	Share of incidents responded to within the severity-based SLA
Average monthly incidents	Incidents divided by the months with data (12 for full years, 9 for 2026 year to date)
External vs internal	Share by Attack_Origin

6. Dashboard design

The dashboard follows a dark, radial analytics style. The left panel holds the year slicer (2020 to 2026, with 2026 marked as year to date), the headline number, the monthly sparkline and the severity breakdown. The right panel holds average monthly incidents, monthly mean time to respond and the external/internal split. The centre changes with the four navigation tabs. Hovering shows tooltips; the year slicer updates every visual. For 2026 the sparkline and monthly charts stop at September.

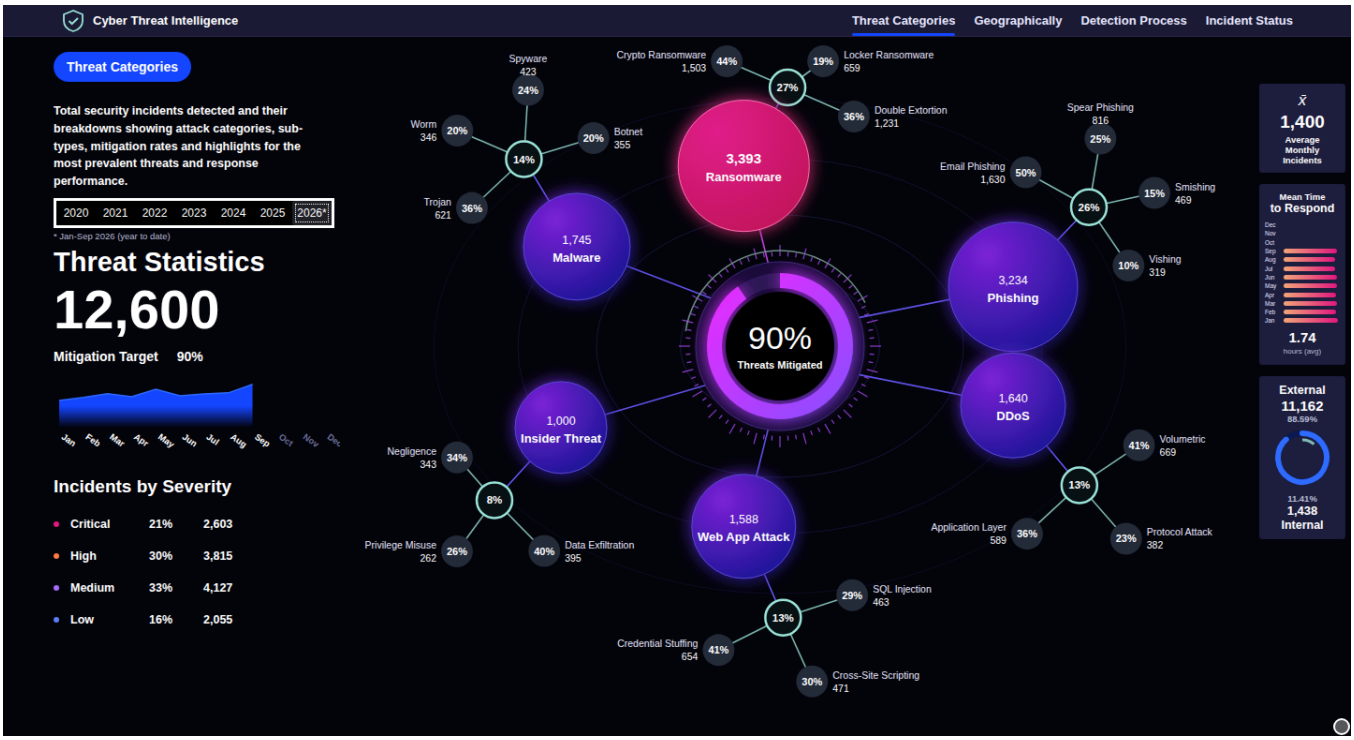


Figure 1 - Threat Categories (2026 YTD): bubbles = categories, hubs = share of total, small nodes = sub-types, centre ring = mitigation rate.



Figure 2 - Geographic view (2026 YTD): bubble size by source country, with ranked countries and targeted industries.

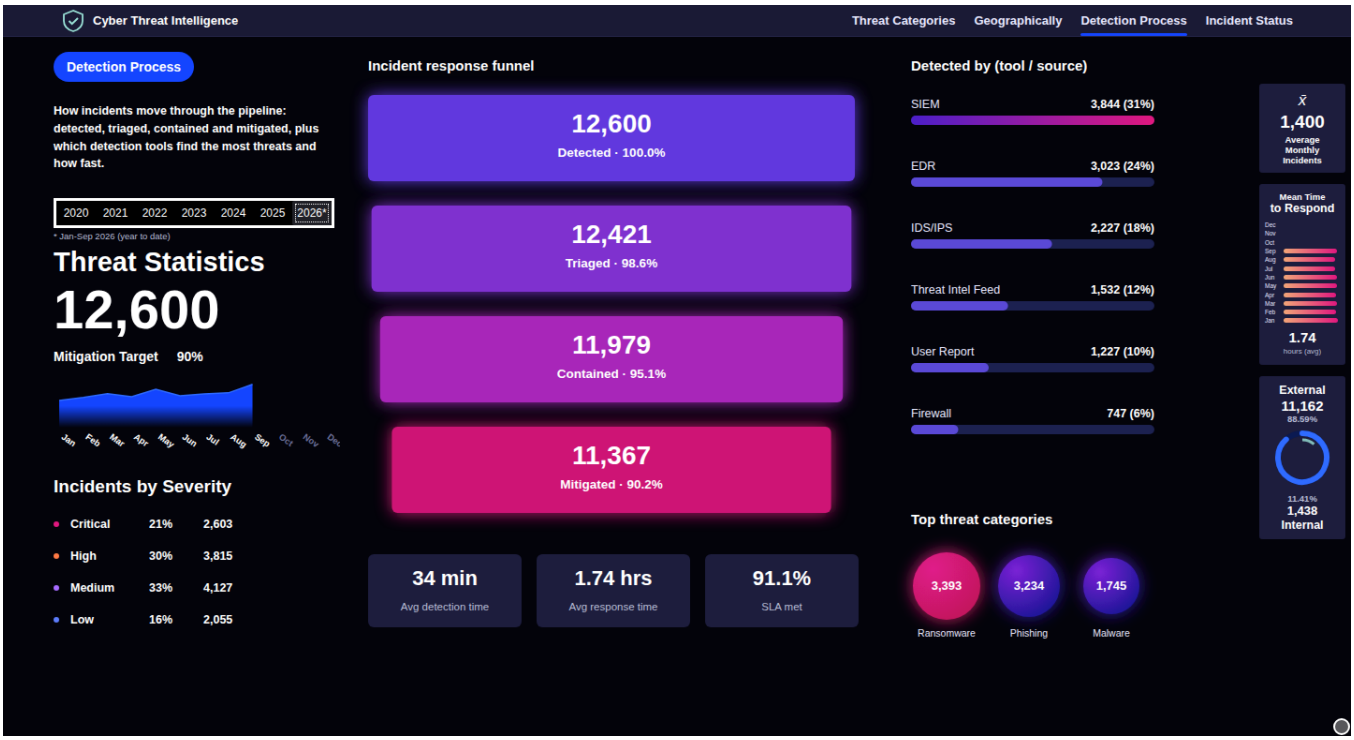


Figure 3 - Detection Process (2026 YTD): response funnel, detection sources and key timing KPIs.



Figure 4 - Incident Status (2026 YTD): stacked monthly status and the severity-by-status grid.

7. Key insights

- **Volume is rising fast.** Incidents grew from 7,200 (2020) to 16,000 (2025), +122%. The average month went from 600 to 1,333, and to 1,400 in 2026 year to date. December is the busiest month in every full year: across the six Decembers 2020 to 2025 there were 6,401 incidents against about 5,633 in an average month (14% above).
- **Ransomware is the fastest-growing threat and is now the largest.** Its share went from 12% to 27% (887 incidents in 2020, 3,964 in 2025). In 2026 year to date it leads with 3,393 incidents against 3,234 for phishing. It also carries the highest estimated impact of any category (about $6.7B of the $10.4B total, 64%, ahead of Malware at $951M).
- **Phishing is still close behind.** It was the largest category in every year to 2025 and is 26% of 2026 year-to-date incidents; email phishing is roughly half of it, so user awareness and mail filtering remain high-value controls.
- **Response is getting faster.** Mean time to respond fell 46% (from 3.21 to 1.74 hours) and SLA compliance rose from 65% to 91%.
- **The mitigation target is now met, narrowly.** The mitigation rate improved from 80.0% to 90.2%. It stayed below the 90% target in every full year (88.5% in 2025) and reaches it for the first time in 2026 year to date. Insider Threat has the lowest category mitigation rate overall (85.0%). Incidents detected in recent weeks have had less time to be resolved, so the year-to-date rate should be re-checked at year end.
- **Attack origin is concentrated.** China and Russia together account for 35% of all incidents in this dataset; Finance and Healthcare are the most targeted industries.
- **Critical incidents are a growing slice:** 16% of incidents in 2020, 20% in 2025 and 21% in 2026 year to date; in absolute terms critical incidents rose 2.8 times between 2020 and 2025 (1,157 to 3,221).
- **Detection relies on a few tools.** SIEM and EDR detect 54% of all incidents.

8. Recommendations

- Prioritise ransomware readiness: tested offline backups, EDR coverage and an incident playbook, because it is now the largest, highest-impact and fastest-growing category.
- Add staffing or automation (SOAR) before the autumn peak: incidents run above average from September and December is about 14% above an average month.
- Keep investing in phishing awareness and email security, which still produce almost as many incidents as ransomware.
- Review the open and under-investigation backlog weekly, starting with Critical and High severity, to hold the mitigation rate at or above the 90% target (179 incidents were still open in 2026 year to date).
- Add a second detection layer for the incident types that SIEM and EDR find least often to reduce reliance on two tools.

9. Limitations and future work

- The data is synthetic, so the numbers demonstrate the analysis method, not real-world rates.
- 2026 is a partial year (January to September). Use rates and averages, not totals, to compare it with full years.
- The interactive dashboard is delivered as a self-contained HTML file (HTML, CSS, JavaScript and SVG). The same measures and layout can be rebuilt in Power BI from the processed CSV or the Excel workbook; the SQL file holds the equivalent queries.
- Future work: connect to a live SIEM export, add forecasting of monthly incidents, add MITRE ATT&CK technique mapping and alerting when the mitigation rate drops below target.

10. Tools used

Python (pandas, NumPy) for data generation and cleaning; SQL for KPI queries (tested on SQLite); Excel (openpyxl, formulas) for the summary workbook; HTML/CSS/JavaScript/SVG for the interactive dashboard; Playwright for screenshots and the interaction video.